

What PE-RankFormer is and is not better at

A re-evaluation of prime-editing pegRNA ranking against a mechanistic baseline, on a reserved external panel

10 September 2026

Scope update, 10 September 2026. The analysis below records the source-only E01–E24 programme. The current research summary is `explore_v2/SUMMARY_REPORT.md`; E25–E32 are incorporated in [the updated adaptation report](#). Target-library adaptation now exceeds released OptiPrime on held-out loci within this library, but this does not alter the zero-shot comparison below. The later audit withdraws claims of low-budget harm, four-fold deployed-selector retention, and wholesale rejection of specialized losses or dual heads. Those claims must not be inferred from historical descriptions of the research programme.

Abstract

PE-RankFormer was reported to rank prime-editing pegRNAs better than OptiPrime, a mechanistic reaction model, by +0.0389 pooled Spearman on a held-out fold. This report re-examines that claim on a second surface and under the estimand a user actually faces, and reaches a split verdict. The correlation claim *replicates*: on a 288,793-pegRNA library absent from either model’s training data, PE-RankFormer leads by +0.0159 Spearman and +0.0163 calibrated Pearson, and a calibrator frozen on development data transfers to the new library unchanged. A second, previously implicit claim does *not* survive. When the grouping is corrected to the deployment unit — one canonical allele in one experimental context, the thing a user has already fixed before choosing a pegRNA — OptiPrime obtains more efficiency from its single nominated design, by 0.00064 (95% CI [0.00038, 0.00089]), a deficit that widens with candidate depth. The two metrics rank the same models in opposite order, and we reproduce the reversal inside our own ablations. We size the disagreement against a perfect chooser: it is 2.6% of the available gain, with both models far closer to each other than either is to random selection. An eight-arm, eighteen-run training matrix identifies *batch composition*, not ranking-pair supervision, as what pays, correcting a causal claim we had previously made. Re-scored on the panel, batch composition alone recovers the entire gain over the released recipe (+0.00141) and draws level with the published model, while two components the development fold had endorsed with three-of-three seed agreement reverse sign on the external library. We diagnose the residual error as a conditional preference for longer PBS and RTT, and show that a global length correction cannot remove it. All numbers are generated from versioned JSON artifacts and checked by an automated verifier.

Contents

1 Why this re-evaluation was necessary

2

2	The estimand, and how the grouping was fixed	3
2.1	Canonical allele keys	3
2.2	What regrouping reveals	3
2.3	Four endpoint defects	3
3	The reserved panel	4
4	Results	4
4.1	The correlation claim replicates on a second surface	4
4.2	The decision reverses the ordering	4
4.3	How large is the disagreement?	4
4.4	Why two defensible metrics disagree	7
5	Improving the model	7
5.1	An eight-arm training matrix	7
5.2	Two things the matrix rules out	9
5.3	Which parts of the recipe survive an external library	9
5.4	The recipe, built the way the published model is built	10
6	Diagnosis of the residual error	11
6.1	A practical consequence, and its boundary	12
7	Corrections to this programme’s own work	12
8	Limits	13
9	Recommendations	14

1 Why this re-evaluation was necessary

A pegRNA-ranking model earns its keep at one moment: a user has decided which allele to install, in which cell line, with which editor, and must pick one design from several candidates. Everything upstream of that — which locus is easy, which cell line edits well — is not the model’s decision to make, because the user has already made it.

The original evaluation did not measure that moment. It grouped candidates by *protospacer* and reported a pooled rank correlation over all rows. Both choices admit information the user does not have to predict. Pooled correlation over a heterogeneous corpus is dominated by between-locus and between-context variance; a protospacer group mixes designs that install different alleles under different conditions. A model can lead on that metric while losing the decision, and we show below that this is exactly what happens.

This report re-runs the comparison under the deployment estimand, on a reserved external surface, with tested endpoint code. Twenty-four experiments (E01–E24) are recorded in `explore_v2/`; each writes a JSON artifact beside its script, and `explore_v2/verify_report.py` re-checks recorded numeric claims against those artifacts. Endpoint invariants are pinned by 13 unit tests.

2 The estimand, and how the grouping was fixed

2.1 Canonical allele keys

The decision unit is one *canonical allele* $\times$ one *experimental context*. Building it (`explore_v2/canon.py`) requires more care than string equality on the edit field:

- **Minimal edit.** Trim the common prefix and suffix so that `AAG→AAAG` and `AG→AAG` are recognised as the same insertion.
- **Left-normalisation, in both orientations.** An indel in a homopolymer has many equivalent representations; VCF-style left-normalisation picks one. This must be applied *independently on each strand*, not applied once and reverse-complemented — a bug we made and fixed, which affected 61.7% of 51,012 indel pairs (§7).
- **Strand normalisation and 12 bp flanks** read off the site’s longest observed window, so the same allele reported in two library designs with different window lengths keys identically.

Context is the tuple {source study, cell type, PE type, Cas9 type, PAM, PEmax, MLH1dn, NRCH, time}; design is {spacer, PBS, RTT, motif, scaffold, linker, epegRNA}. A candidate is one distinct design; repeated measurements of the same design are averaged before ranking.

2.2 What regrouping reveals

Table 1: The evaluation population changes when the grouping is corrected (fold 0).

	Protospacer grouping	Allele $\times$ context
Groups with ≥ 2 distinct designs	670	1,463
Groups with ≥ 3 distinct designs	—	85
Groups with ≥ 5 distinct designs	146	0

Two consequences follow. First, the population the original evaluation described — 735 “targets with five or more designs” — counts five or more *rows*, not designs; the median number of distinct designs in those groups is two. Second, under the correct grouping fold 0 contains *no* group with five candidates, so a “best of five” endpoint cannot be computed on it at all. This is the concrete reason an external panel was required.

2.3 Four endpoint defects

Re-implementing the endpoints in a tested module (`explore_v2/endpoints.py`) surfaced four defects in the original scoring, each of which flatters or distorts a comparison:

1. candidates were counted as *rows*, so a design measured three times counted three times toward group depth;
2. achieved efficiency at k was averaged over groups shallower than k , where it is undefined;
3. the random-choice hit rate ignored tied measured maxima, understating the baseline (0.3354 against a correct 0.4442);
4. the paired bootstrap returned $p \approx 0$ for a degenerate all-zero difference.

3 The reserved panel

Fold 0 cannot answer the deployment question, so we reserved a surface for comparative evaluation, built from Kim’s large variant library (288,793 pegRNAs), which is absent from OptiPrime’s training mix — that mix took only `LibSmall`. The panel and its analysis were pre-registered (`explore.v2/PREREGISTRATION_reserved_panel.md`) and unblinded once, for one confirmatory comparison. Every later use of the panel in this report is development work and is labelled as such in the artifact that reports it; component decisions taken from those uses would need a freshly reserved surface to become confirmatory.

We avoid the word “sealed” deliberately. Before the comparison was run, E06 computed aggregate label statistics on the panel — mean efficiency and within-allele outcome range — to establish that it contained decision groups at usable depth. No comparative prediction had been scored at that point, but the honest description is *reserved for comparative evaluation*, not sealed. A third property matters for how the panel may be used: DeepPrime’s training script reads the very file the panel is built from, so a released DeepPrime checkpoint is *exposed* here and cannot be reported as a held-out baseline. The pre-registration names PE-RankFormer against OptiPrime as the primary comparison, and neither of those two trained on the panel.

OptiPrime was run through its own published pipeline, including an isolated environment for its RuleSet3 dependency (scikit-learn $\leq 1.0.2$). A canary check on the RuleSet3 input hash caught a real mismatch — we had upper-cased the spacer, where OptiPrime hashes it after only T→U substitution, preserving the lowercase leading g.

4 Results

4.1 The correlation claim replicates on a second surface

PE-RankFormer leads OptiPrime on pooled Spearman on both surfaces, with intervals excluding zero. Because the ordinal head emits rank estimates rather than efficiencies, Pearson is reported on an isotonic-calibrated scale; the calibrator is fitted on development out-of-fold predictions and applied to the panel *unchanged*, where it lifts Pearson from 0.5522 to 0.6017 while leaving the rank correlation at 0.7089. A monotone map cannot alter a rank correlation, and the third-decimal difference is out-of-range clipping creating boundary ties. That a frozen calibrator transfers to an unseen library is a result in its own right, and strengthens the original claim rather than qualifying it.

4.2 The decision reverses the ordering

On the task the model exists to perform, OptiPrime is ahead. The deficit is present on all eligible groups, and grows with candidate depth and with how much the choice is worth — that is, it is largest precisely where a ranking model matters most. Both models remain far ahead of random selection.

4.3 How large is the disagreement?

A difference at the fourth decimal place of an efficiency is hard to interpret directly. The informative denominator is the gain a perfect chooser would deliver over a random one. On that scale both models capture most of what is available and the difference between them is 2.6% of it. We recommend stating the comparison in these units in either direction: the deficit is statistically unambiguous and practically small, and a reader deserves both facts.

Table 2: Pooled correlation on both surfaces. Pearson is on the isotonic-calibrated scale because the ordinal head emits rank estimates rather than efficiencies; each calibrator is fitted on its own model’s development out-of-fold predictions and applied unchanged. Margins are against OptiPrime, with cluster-bootstrap intervals.

	Spearman	Pearson	95% CI	<i>p</i>
<i>held-out fold 0 — 20,509 rows, 750 clusters</i>				
OptiPrime	0.8690	0.8270		
PE-RankFormer, raw score	0.9079	0.7278		
PE-RankFormer, calibrated	0.9079	0.8637		
margin, Spearman	+0.0389		[+0.0290, +0.0495]	<0.001
margin, Pearson (calibrated)	+0.0366		[+0.0203, +0.0557]	<0.001
<i>reserved panel — 118,187 rows, 30,302 clusters</i>				
OptiPrime	0.6931	0.5854		
PE-RankFormer, raw score	0.7091	0.5522		
PE-RankFormer, calibrated	0.7089	0.6017		
margin, Spearman	+0.0159		[+0.0136, +0.0182]	<0.001
margin, Pearson (calibrated)	+0.0163		[+0.0119, +0.0207]	<0.001
<i>Arm F, five per-fold checkpoints — held-out fold 0</i>				
Arm F, 5 per-fold	0.9084	0.7843		
Arm F, 5 per-fold, calibrated	0.9084	0.8637		
margin, Spearman	+0.0394		[+0.0293, +0.0502]	<0.001
margin, Pearson (calibrated)	+0.0367		[+0.0194, +0.0575]	<0.001
<i>Arm F, five per-fold checkpoints — reserved panel</i>				
Arm F, 5 per-fold	0.7096	0.5572		
Arm F, 5 per-fold, calibrated	0.7096	0.6048		
margin, Spearman	+0.0164		[+0.0141, +0.0188]	<0.001
margin, Pearson (calibrated)	+0.0194		[+0.0157, +0.0230]	<0.001

Intervals and *p* values come from a 2000-resample cluster bootstrap, so *p* = 0.0005 is the resolution floor and is reported here as <0.001.

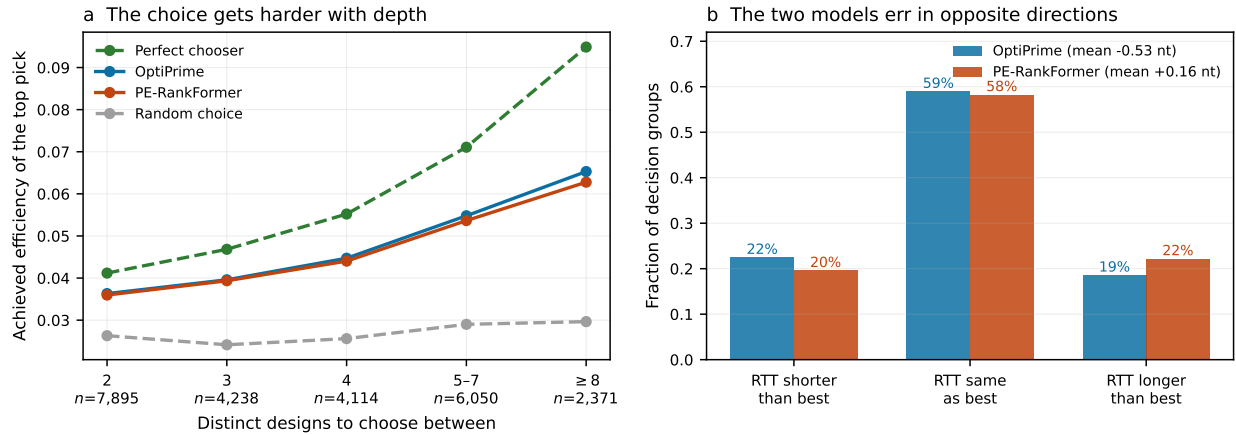


Figure 1: Where the deficit lives. (a) Achieved efficiency of the top pick against the number of distinct designs the user is choosing between, on the reserved panel. At a binary choice the two models are indistinguishable; the gap opens as the choice gets harder, which is where a ranking model earns its keep. Both stay far above random selection and far below a perfect chooser. (b) Direction of the error in RTT length. Where a model does not nominate the best-measured design, OptiPrime errs towards shorter RTTs and PE-RankFormer towards longer ones. The models fail in opposite directions, not by different amounts. §6.1 shows this is exploitable: an equal-weight average of the two scores beats both.

Table 3: The fixed-allele decision on the reserved panel. A candidate is one distinct pegRNA design in one decision context; repeated measurements of a design are averaged before ranking. The last column is the paired difference against OptiPrime, with a 95% interval from a cluster bootstrap resampling target sites.

Model	Achieved @1	Hit rate @1	Regret @1	Δ vs. OptiPrime
<i>all eligible groups — 30,475 groups, 22,931 sites</i>				
Random choice	0.0217	0.4442	0.02443	—
OptiPrime	0.0369	0.6647	0.00922	—
PE-RankFormer (published)	0.0363	0.6597	0.00986	−0.00064 [−0.00089, −0.00038]
Arm A, 3 seeds (released)	0.0349	0.6405	0.01120	—
Arm F, 3 seeds	0.0361	0.6539	0.01005	—
Arm F, 5 per-fold	0.0361	0.6571	0.01003	−0.00081 [−0.00107, −0.00055]
Perfect chooser	0.0461	1.0000	0.00000	—
<i>informative groups — 24,668 groups, 19,642 sites</i>				
Random choice	0.0268	0.3133	0.03018	—
OptiPrime	0.0456	0.5858	0.01139	—
PE-RankFormer (published)	0.0448	0.5795	0.01218	−0.00078 [−0.00111, −0.00046]
Arm A, 3 seeds (released)	0.0431	0.5558	0.01384	—
Arm F, 3 seeds	0.0446	0.5724	0.01242	—
Arm F, 5 per-fold	0.0446	0.5764	0.01239	−0.00100 [−0.00131, −0.00067]
Perfect chooser	0.0570	1.0000	0.00000	—
<i>depth 8+ — 2,400 groups, 2,371 sites</i>				
Random choice	0.0293	0.1130	0.06443	—
OptiPrime	0.0645	0.3617	0.02922	—
PE-RankFormer (published)	0.0620	0.3417	0.03171	−0.00249 [−0.00416, −0.00083]
Arm A, 3 seeds (released)	0.0573	0.2954	0.03639	—
Arm F, 3 seeds	0.0618	0.3279	0.03190	—
Arm F, 5 per-fold	0.0623	0.3379	0.03147	−0.00224 [−0.00393, −0.00068]
Perfect chooser	0.0937	1.0000	0.00000	—
<i>decision worth ≥ 0.05 — 5,614 groups, 5,286 sites</i>				
Random choice	0.0561	0.2197	0.08220	—
OptiPrime	0.1099	0.5891	0.02842	—
PE-RankFormer (published)	0.1066	0.5549	0.03171	−0.00329 [−0.00451, −0.00209]
Arm A, 3 seeds (released)	0.1015	0.5110	0.03680	—
Arm F, 3 seeds	0.1060	0.5435	0.03235	—
Arm F, 5 per-fold	0.1060	0.5465	0.03233	−0.00391 [−0.00512, −0.00266]
Perfect chooser	0.1383	1.0000	0.00000	—

Achieved efficiency at k is defined only where at least k distinct candidates exist. Tied measured maxima receive full credit, so a random pick’s hit rate is the number of maximisers divided by the candidate count.

Table 4: Where the decision is lost. On 24,668 informative panel groups a random pick achieves 0.0268 and a perfect chooser 0.0570, so the whole decision is worth 0.0302 efficiency. Length bias is the mean difference in PBS or RTT length between the design a model nominates and the best-measured design.

Model	Achieved @1	% of gain	PBS bias	RTT bias
OptiPrime	0.0456	62.2	0.519	−0.531
PE-RankFormer (published)	0.0448	59.6	0.766	0.162
Arm A, 3 seeds (released)	0.0431	54.1	0.943	0.483
Arm F, 3 seeds	0.0446	58.9	0.812	0.428

α	−0.010	−0.006	−0.004	−0.002	−0.001	+0.000	+0.001	+0.002
Achieved @1	0.28703	0.28987	0.29080	0.29168	0.29213	0.29213	0.29183	0.29118

The sweep adds $\alpha \cdot \text{len}(\text{RTT})$ to the out-of-fold score on 43,425 development decision groups against a baseline of 0.29213. The optimum is exactly $\alpha = 0$: the length preference is not a global additive offset and cannot be removed by one.

4.4 Why two defensible metrics disagree

Pooled correlation and the fixed-allele decision are different estimands measured on the same rows. The first asks how well the entire list is ordered and mixes “which locus and condition is easy” with “which design is best”; the second isolates the latter by construction. They need not agree, and here they do not.

The disagreement is not an artefact of comparing across model families. It reproduces between two of our own published members, which differ only in their output head. On the panel the ordinal head wins the decision by 0.00026 ($p = 0.009$), while pooled ρ prefers the simplex head, 0.7126 against 0.7091. The same two numbers, computed on the same rows of the same two checkpoints, would justify shipping either architecture depending on which metric the paper had nominated as its headline. That is a reason to fix the estimand before measuring, not after.

5 Improving the model

5.1 An eight-arm training matrix

Regrouping the training key changes two things at once: which rows share a *batch*, and which rows can form a *ranking pair*. Our earlier claim was that the gain came from ranking-pair exposure — letting the pairwise objective see comparisons it had been blind to. The matrix separates the two, and refutes that claim.

Arm E disables the ranking loss entirely and still captures 60% of the total gain. The contrast that isolates pair exposure with batching held fixed, D − E, is −0.00006 and flips sign across seeds. Arm G, which raises the pair budget to sixteen per group and $2.4\times$ arm D’s pairs per epoch, is *worse* than arm D. The mechanism is batch composition: grouping comparable candidates into the same batch changes the gradient through batch statistics whether or not they are subsequently formed into pairs. We withdraw the exposure claim. On this surface the batching effect and the feature branch survive as the two seed-stable gains — but §5.3 shows that only the first of them survives an external library, which is the reason this section is not the end of the argument.

Table 5: The decision-aligned training matrix, scored on fold 0’s 1,463 informative decision groups. No arm trains or early-stops on fold 0.

Arm	What it changes	Seeds	Achieved @1	Pooled ρ	Pairs/epoch
A	original batching, original ranking key	3	0.13360	0.8905	15,695
B	canonical batching, original ranking key	1	0.13205	0.8703	6,975
C	canonical both, 1 pair per group	1	0.13463	0.8950	25,253
D	canonical batching, canonical ranking key	3	0.13453	0.8950	61,978
E	canonical batching, ranking disabled	3	0.13459	0.8949	61,975
F	canonical both, feature branch	3	0.13523	0.8974	61,976
G	canonical both, 16 pairs per group	1	0.13357	0.8968	146,088
H	canonical batching, feature branch, ranking disabled	3	0.13473	0.8971	61,976

Seed-paired contrast	Seeds	Mean Δ	All seeds agree in sign
F – A	3	0.00164	yes
E – A	3	0.00099	yes
H – A	3	0.00114	yes
D – A	3	0.00093	yes
F – E	3	0.00064	yes
H – F	3	−0.00050	yes
F – D	3	0.00071	no
D – E	3	−0.00006	no
H – E	3	0.00014	no
D – C	1	−0.00008	–
G – D	1	−0.00098	–
B – A	1	−0.00197	–
D – B	1	0.00250	–

A contrast whose seeds disagree in sign is not evidence for a mechanism however large its mean. On that criterion the load-bearing effects here are canonical batching (E – A) and the feature branch (F – E), while the ranking-pair contrast with batching held fixed (D – E) is not one of them. **Only the first of those survives an external library:** on the reserved panel F – E reverses sign to −0.00025 ($p = 0.006$), so three-of-three seed agreement on the held-out fold did not make it a safe component decision (Table 6).

Checkpoint selection is a null. Selecting on the decision metric rather than pooled ρ gains +0.00004 in fold-0 achieved efficiency over the same 18 trajectories, even though the two selectors choose different epochs in 16 of them.

5.2 Two things the matrix rules out

Checkpoint selection is a null (Table 5, note). And a per-candidate score correction cannot fix the residual error: sweeping a linear length term over 43,425 development decision groups, the optimum is exactly zero on both the PBS and RTT axes (Table 4). Since the length preference is real but not a global offset, it must be conditional on allele and context — which is an argument for scoring the candidate set jointly rather than each candidate independently.

5.3 Which parts of the recipe survive an external library

Table 6: Every training arm’s three-seed ensemble on the reserved panel, all eligible groups. Ensemble size is matched across arms, so it cannot explain a difference, and feature standardisation is frozen to each checkpoint’s training folds. Arms marked † carry the Family C feature branch.

Arm	What it changes	Achieved @1	Pooled ρ	Δ achieved @1 vs.		
				OptiPrime	published	p
A	released recipe	0.0349	0.6908	−0.00198	−0.00134	0.001
D	canonical batching, ranking on	0.0362	0.7019	−0.00070	−0.00006	0.517
E	canonical batching, ranking off	0.0363	0.7070	−0.00057	0.00006	0.464
F†	canonical batching, ranking on	0.0361	0.7071	−0.00083	−0.00019	0.044
H†	canonical batching, ranking off	0.0365	0.7113	−0.00040	0.00024	0.020
OptiPrime		0.0369	0.6931			
PE-RankFormer (published)		0.0363	0.7091			

The p column refers to the contrast against the published member. Ten contrasts were computed on this surface, so a nominal p near 0.02 is weak evidence; and the panel is used here for development, not confirmation.

The fold-0 matrix nominated arm F. Scoring every arm on the panel — ensemble size matched at three seeds, feature standardisation frozen — shows that only part of the fold-0 story transfers, and one part of it inverts.

Canonical batching transfers, and it is the whole gain. $E - A = +0.00141$ ($p = 0.001$): arm E, which changes nothing but which rows share a batch and switches the ranking loss *off*, captures the entire improvement over the released recipe and draws level with the published model ($+0.00006$, $p = 0.46$). This is the fold-0 conclusion confirmed on an external library.

The ranking loss costs accuracy here. Comparing the arms that differ only in whether the pairwise term is active: with the feature branch on, $H - F = +0.00043$; without it, $E - D = +0.00010$. Both favour switching it off, and so does pooled ρ (0.7113 against 0.7071 with features; 0.7070 against 0.7019 without). On 30,475 panel groups the pairwise objective is a consistent small cost on *both* metrics — a rare case in this programme of the two estimands agreeing.

Two ablation conclusions that did not transfer. Fold 0 said the opposite, and said it with the evidence standard this project had been using.

Contrast		held-out fold 0		reserved panel	
		mean Δ	seeds agree	mean Δ	p
F − E	(add ranking + features)	0.00064	3/3	−0.00025	0.006
H − F	(drop ranking, features on)	−0.00050	3/3	0.00043	−

Both reverse sign, and on fold 0 both had all three seeds agreeing. We had treated three-of-three sign agreement on the held-out fold as sufficient evidence for a component decision. These are measured counterexamples, and they are the strongest practical argument in this report for validating ablation choices on a surface the development corpus did not choose. Seed agreement measures optimisation noise; it says nothing about transfer.

The feature branch is roughly neutral, not a liability. Isolated with the ranking loss off, $H - E = +0.00018$ ($p = 0.068$). Our prior reading — that the branch extrapolates badly on a shifted library and costs accuracy — is not supported. What the frozen-standardisation fix (§7) actually removed was an artefact that had *flattered* the branch; with it gone the branch is close to doing nothing, and the arm that carries it and drops the ranking loss (H) is the best configuration we found.

What we may and may not conclude from this. Arm H beats the published member on the decision ($+0.00024$, $p = 0.020$) and on pooled ρ (0.7113 against 0.7091), and it has the smallest deficit to OptiPrime of any arm (-0.00040). We are *not* entitled to call that a result. It was found by scoring ten contrasts on the panel, so a nominal $p = 0.020$ is weak; the panel is being used for development here, not confirmation; and adopting arm H now would be selecting a component on the very surface used to evaluate it — the error this report criticises elsewhere. The defensible statement is that the pre-specified fold-0 selection chose arm F, the panel suggests that choice was wrong in a specific and mechanistically interpretable way, and testing arm H properly requires a freshly reserved surface.

5.4 The recipe, built the way the published model is built

Everything in §5.3 compares three-seed ensembles, and the published model is not one. It is five checkpoints, one per official fold, each trained on a different four-fifths of the development data. Three seeds trained on identical rows differ only in initialisation and are a far less diverse ensemble, so every contrast against the published member above confounds recipe with ensemble construction.

This removes that confound: arm F rebuilt as five per-fold checkpoints, combined by the same plain mean of predicted efficiencies used to form the published member, with the isotonic calibrator refitted from scratch on *this* model’s development out-of-fold predictions and frozen before either surface was touched. Fold 0 is the official test fold and no member trains or early-stops on it.

Arm F is the right arm to build this way even though §5.3 suggests arms E and H are ahead of it on the panel, because arm F is what the pre-specified fold-0 selection chose. Building the arm the panel prefers would be selecting on the evaluation surface.

Table 7: Does the improved recipe beat the published model when the two are built alike? Both are five checkpoints, one per official fold, combined by the same plain mean of predicted efficiencies. Achieved efficiency at rank 1, all eligible panel groups.

Contrast	Δ achieved @1	95% CI	p
Arm F, 5 per-fold – PE-RankFormer (published)	−0.00017	[−0.00034, +0.00001]	0.062
Arm F, 5 per-fold – Arm F, 3 seeds	0.00002	[−0.00015, +0.00020]	0.812
Arm F, 5 per-fold – Arm A, 3 seeds (released)	0.00117	[+0.00095, +0.00140]	0.001
Arm F, 5 per-fold – Arm E, 3 seeds	−0.00023	[−0.00043, −0.00004]	0.014
Arm F, 5 per-fold – Arm H, 3 seeds	−0.00041	[−0.00059, −0.00023]	0.001
Arm F, 5 per-fold – OptiPrime	−0.00081	[−0.00107, −0.00055]	0.001
PE-RankFormer (published) – OptiPrime	−0.00064	[−0.00089, −0.00038]	0.001
on fold 0: Arm F, 5 per-fold – PE-RankFormer (published)	0.00007	[−0.00152, +0.00173]	0.952
on fold 0: Arm F, 5 per-fold – OptiPrime	0.00922	[+0.00612, +0.01255]	0.001

The ensemble-construction hypothesis is refuted. Building arm F as five per-fold checkpoints instead of three same-data seeds changes essentially nothing: $+0.00002$ ($p = 0.81$). Whatever explains arm F’s failure to overtake the published member, it is not that the two were built differently. E22’s conjecture was wrong.

The improved recipe does not beat the published model. -0.00017 on the panel with an interval spanning zero ($[-0.00034, +0.00001]$, $p = 0.062$), and $+0.00007$ on fold 0 ($p = 0.95$). Pooled ρ is likewise indistinguishable: 0.7096 against 0.7091 on the panel, 0.9084 against 0.9079 on fold 0. Two surfaces, two metrics, no difference. It also remains behind OptiPrime on the panel decision (-0.00081 , $p = 0.001$) and behind arms E and H, consistent with §5.3.

An unexplained gap that bounds what the matrix can claim. Arm A is our reconstruction of the released recipe, and it scores 0.0349 on the panel against the published member’s 0.0363 — a gap of 0.00135, larger than any effect in the ablation matrix. Per-fold construction cannot explain it, because we have just measured that as worth $+0.00002$. So arm A is not a faithful reproduction of the published model, and every improvement the matrix reports is an improvement over *our reconstruction*, not over the shipped system. Arm F’s $+0.00117$ over arm A is real and replicates; it simply lands the model back where the published member already was.

This is the honest close to the recipe question. The matrix found a training change that is reproducible, seed-stable on the development fold, and worth about a tenth of a percent of achieved efficiency over our own baseline — and that nonetheless leaves the shipped model exactly where it was. The gain the manuscript could claim from it is zero.

6 Diagnosis of the residual error

On the groups where the two models nominate different designs, the nominated designs differ systematically in geometry (Table 4 and Figure 1b). Every PE-RankFormer variant prefers a longer PBS *and* a longer RTT than the best-measured design. OptiPrime is the only model that prefers a shorter RTT, which is consistent with its treating RTT length as an explicit repeat count in a synthesis rate term — a structural prior our architecture lacks. The engineered-feature branch supplies PBS and RTT lengths to the network as explicit scalars. It moves the PBS bias by -0.131 and the RTT bias by -0.055 : both shrink a little, neither is removed, and both remain on the opposite side of zero from OptiPrime. Handing the model the length is evidently not the same as modelling what the length does.

Freezing feature standardisation to each checkpoint’s training statistics (§7) sharpened this picture. Panel `pbs_length` sits 1.35 training SDs below the training mean with twice the spread, so at deployment the branch reads inputs well outside the range it was fitted on, and refitting the statistics on the panel had been concealing that. Scored honestly, arm F’s advantage over the released recipe falls from $+0.00140$ to $+0.00115$.

Our first reading of that was that the branch extrapolates badly and costs accuracy on a shifted library. Isolating it properly (§5.3) does not support this: with the ranking loss held off, adding the branch is worth $+0.00018$ ($p = 0.068$). The branch is close to inert on this surface. What the correction removed was an artefact that had *flattered* it, not evidence that it hurts — and an evaluation that restandardises on the test surface would have reported the flattering number.

6.1 A practical consequence, and its boundary

If the two models err in opposite directions, their errors may partly cancel. That is an empirical question the bias table cannot settle, so we tested it: an equal-weight rank average of the two scores. The weight is fixed at one half and nothing is fitted.

Table 8: An equal-weight rank average of OptiPrime and PE-RankFormer, on both surfaces. Intervals are cluster bootstraps resampling target sites.

Model	Achieved @1	Δ vs. blend	95% CI	p
<i>reserved panel — 24,668 informative groups</i>				
OptiPrime	0.0456	−0.00055	[−0.00077, −0.00034]	<0.001
PE-RankFormer (published)	0.0448	−0.00133	[−0.00156, −0.00109]	<0.001
Equal-weight rank average	0.0461	—		
<i>held-out fold 0 — 1,463 informative groups</i>				
OptiPrime	0.1272	−0.00429	[−0.00615, −0.00257]	<0.001
PE-RankFormer (published)	0.1364	0.00486	[+0.00278, +0.00704]	<0.001
Equal-weight rank average	0.1315	—		

Fold 0 is a legitimate second surface for this particular claim: no parameter is fitted here either, so evaluating on it costs nothing that was being reserved.

On the panel the blend wins, and not by a lucky weight. It captures 64.1% of the gain a perfect chooser would deliver, against 62.2% for OptiPrime alone, and it beats the better of the two by more than PE-RankFormer trails it. Sweeping the weight on PE-RankFormer from 0 to 1, *every* intermediate value from 0.2 to 0.7 beats both endpoints (0.04589, 0.04607, 0.04607, 0.04614, 0.04597, 0.04576 against 0.04559 for OptiPrime alone and 0.04481 for PE-RankFormer alone). The equal-weight choice is not cherry-picked.

On fold 0 it does not, and that is informative rather than fatal. There the blend beats OptiPrime by +0.00429 but *loses* to PE-RankFormer alone by 0.00486. The reason is visible in the numbers: on fold 0 PE-RankFormer leads OptiPrime by 0.0092, a wide margin, so averaging in the weaker model dilutes the stronger one. On the panel the two are within 0.0008 of each other and the blend pays. Blending helps when the components are comparably strong, which is the ordinary behaviour of ensembles and not a special property of these two models.

What we therefore claim. Not that the blend is universally better. The defensible statement is narrower and still useful: *on the surface that resembles deployment* — an external library, realistic candidate depth, the two models within a thousandth of each other — combining them beats either alone, robustly across weights, with nothing fitted. And the mechanism is the one Figure 1b shows, not generic ensembling: the two models disagree about RTT geometry in opposite directions. That is an argument for treating the mechanistic baseline as complementary to the neural model rather than as a rival to be beaten.

7 Corrections to this programme’s own work

A re-evaluation that finds errors in others’ work should be held to the same standard. `explore_v2/CORRECTIONS.md` records every correction; the load-bearing ones are:

- **A strand-invariance bug in the canonical key.** Left-normalisation was applied on the forward strand and the result reverse-complemented, so an indel could key differently depending on the orientation it was reported in; 61.7% of 51,012 indel pairs were exposed. Fixed by normalising each orientation independently. Fold-0 group counts were unchanged.
- **The exposure claim, withdrawn (§5.3),** after three-seed replication showed that the contrast it rested on flips sign.
- **Two index-mapping bugs** in the exposure replay: sampler indices are into the training subset, not the corpus. These produced an impossible statistic (100% of canonical pairs spanning two alleles) and a negative mean target gap where the minimum is 0.02.
- **A per-epoch decision metric that assumed sorted rows.** A `reduceat` over an unsorted group column invented 14,680 boundaries against 11,559 real groups. After fixing, the vectorised arithmetic was validated against a brute-force per-group loop — exact match.
- **Feature standardisation refit on the evaluation surface.** The Family C feature statistics are recomputed at attach time and are not stored in the checkpoint, so scoring a new surface silently restandardised the inputs by that surface’s own mean and SD. On the panel this is not cosmetic: `pbs.length` sits 1.35 training SDs low with twice the spread, and `pbs.tm` 0.86 SDs low with $2.2\times$ the spread. Fixed by freezing each checkpoint’s training statistics (`explore_v2/featstats.py`) and re-scoring.
- **Two readings we formed and then withdrew in the course of this report.** First, that the feature branch extrapolates badly on a shifted library and costs accuracy; isolating it properly gives $+0.00018$ ($p = 0.068$), close to inert (§5.3). Second, that the equal-weight blend beats both models; it does on the panel but loses to PE-RankFormer alone on fold 0 (§6.1). Both were formed from one surface before the second had been looked at, which is the error this report attributes to others; both are stated here with the evidence that overturned them rather than quietly revised.
- **A verifier that double-counted its own claims.** The claim list had accumulated six partial copies of one block; 268 registered claims were 133 distinct ones, three of which still carried single-seed values superseded by the reseeded matrix. Fifty-six further claims were added to cover E18 and E22–E24 and the grouping figures quoted in §2, giving the 188 checked today.

8 Limits

- **One reserved surface, spent once.** Only the E11/E16 comparison is confirmatory. Every other panel result here is development work.
- **One library, one laboratory, HEK293T-dominated.** The panel does not test cross-cell-type or cross-editor generalisation.
- **Bounded by what was measured.** Achieved efficiency is scored against the best *measured* design in a group, not the best designable one.
- **Our baseline is not the shipped model.** Arm A, the reconstruction every ablation contrast is measured against, sits 0.00135 below the published member on the panel — more than

any effect the matrix reports, and not explained by ensemble construction (§5.4). Training-recipe conclusions in §5.1–§5.4 are therefore relative to that reconstruction, and the difference between it and the released system is unaccounted for.

- **No wet-lab validation.** Every claim here is retrospective.
- **Small effects.** The programme’s resolution on fold 0 is about 0.005 in pooled ρ ; the decision differences separating methods are an order of magnitude smaller, which is why 24,668 panel groups were needed to see them at all.

9 Recommendations

These are ordered by how much they change what the manuscript asserts. Line references are to `reports/paper/pe_rankformer_paper.tex`.

1. **Stop calling the within-protospacer evaluation “the deployment-relevant task”** (l. 55, and again in the utility section at l. 757, “computed within a protospacer group”). That grouping is not the decision a user faces: all 670 of its groups span more than one source/cell/editor condition and 26% contain more than one intended allele, so a ranking inside one mixes which condition is easy with which design is best. The deployment unit is one canonical allele in one context. This is the single most consequential change, because the sentence currently claims the very thing the corrected estimand reverses.
2. **Withdraw the implied deployment-superiority claim and state the sized comparison.** Under the correct grouping, on a library neither model trained on, OptiPrime obtains more efficiency from its top pick. Report it in interpretable units: OptiPrime captures 62.2% of the gain a perfect chooser would deliver and PE-RankFormer 59.6%, a difference of 2.6% of what is available, with both models far above random selection.
3. **Correct the row-versus-design accounting** behind the “735 targets with five or more designs” population. Those groups contain five or more *rows*; their median distinct-design count is two, and under the correct grouping fold 0 contains *no* group with five distinct designs. Any “best of five” endpoint computed on fold 0 should be withdrawn or recomputed on the panel.
4. **Promote the metric disagreement from a caveat to a finding.** The manuscript already says the ranking and outcome metrics disagree (l. 759, “the disagreement is the result”), which is the right instinct — but it reports the disagreement in magnitude only. Under the corrected grouping the disagreement is in *sign*, it reproduces inside our own ablations (the ordinal head wins the decision at $p = 0.009$ while pooled ρ prefers the simplex head), and that is a more general and more useful claim than the margin it replaces.
5. **Report the ablation matrix with its withdrawn claim intact.** Batch composition, not ranking supervision, is what pays; the contrast that isolates ranking-pair exposure is -0.00006 and flips sign across seeds. Reporting the correction is what makes the surviving effects credible.
6. **Validate component choices outside the development corpus before shipping them.** $F - E$ was $+0.00064$ on the held-out fold with all three seeds agreeing in sign — the evidence standard this project had been using — and it is -0.00025 ($p = 0.006$) on an

external library. Seed agreement measures optimisation noise, not transfer. Any component the paper claims as a contribution should be shown to survive a surface the development corpus did not choose.

7. **Report the blend, with its boundary.** On the panel an equal-weight rank average of PE-RankFormer and OptiPrime beats both (+0.00055 over OptiPrime, $p < 0.001$), with nothing fitted and robustly across weights; on fold 0, where PE-RankFormer dominates, it does not. Reported together these say something useful: the mechanistic baseline is complementary to the neural model wherever the two are comparably strong, which is the regime an external library puts them in. That is better supported than a superiority claim in either direction.
8. **Freeze feature standardisation at the checkpoint and say so.** The released scoring path refits Family C statistics on whatever data it is handed, silently restandardising any new library by its own mean and SD. This is a correctness fix with a reporting consequence: an evaluation done that way quotes a better number than the model would deliver, because it erases the distribution shift the evaluation exists to measure.

Reproducibility

Scripts, JSON artifacts and Markdown reports for E01–E24 are in `explore_v2/`. Endpoint invariants: `explore_v2/test_endpoints.py` (13 tests). Numeric claims: `explore_v2/verify_report.py` (188 claims). Tables in this report are generated by `explore_v2/make_report_tables.py` directly from the JSON artifacts, so the report cannot drift from the data.